Akash® Technolabs

Python List Methods

#PythonNotes

Python List/ Tuple Methods

Python List Methods

- Python has a set of built-in methods that you can use on lists

S. No.	Method	Description of method
1	append()	Add an item to the ending of the list.
2	extend()	Add items of one list to another list.
3	insert()	Adds an item at the desired index position.
4	remove()	Remove the primary item from the list with the required value.
5	pop()	Remove the item from the required index within the list, and return it.
6	clear()	Remove all items from the list.
7	index()	Returns 0-based index of the first matched item. Raises ValueError if an item isn't found within the list.
8	count()	Returns the frequency of an element appearing in the list.
9	sort()	Sorts the objects of the list. It uses a compare function if passed as an argument.
10	reverse()	Reverse the order of the list items in place.
11	copy()	Return a shallow copy of the list i.e. two lists share the identical elements via reference.

sort()

- The sort() method sorts the list ascending by default.

demo.py > ...

```python
mylist1 = ["Java","Python","Angular","React"]
mylist2 = [99,58,6,72,36,96]

mylist1.sort()
print(mylist1)
mylist2.sort()
print(mylist2)
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL

```
D:\code\python_code>python demo.py
['Angular', 'Java', 'Python', 'React']
[6, 36, 58, 72, 96, 99]
```


```python
mylist1 = ["Java","Python","Angular","React"]
mylist2 = [99,58,6,72,36,96]

mylist1.sort()
print(mylist1)
mylist2.sort()
print(mylist2)
```

reverse()

- The reverse() method reverses the sorting order of the elements.

demo.py > ...

```python
mylist1 = ["Java","Python",10,"Angular",99,"React"]

mylist1.reverse()
print(mylist1)
```

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```
D:\code\python_code>python demo.py
['React', 99, 'Angular', 10, 'Python', 'Java']
```

```python
mylist1 = ["Java","Python",10,"Angular",99,"React"]

mylist1.reverse()
print(mylist1)
```

Reverse with Sort

demo.py > ...

```python
mylist1 = ["Java","Python","Angular","React"]
mylist2 = [99,58,6,72,36,96]

mylist1.sort()
mylist1.reverse()
print(mylist1)

mylist2.sort()
mylist2.reverse()
print(mylist2)
```

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```
D:\code\python_code>python demo.py
['React', 'Python', 'Java', 'Angular']
[99, 96, 72, 58, 36, 6]
```

```python
mylist1 = ["Java","Python","Angular","React"]
mylist2 = [99,58,6,72,36,96]

mylist1.sort()
mylist1.reverse()
print(mylist1)

mylist2.sort()
mylist2.reverse()
print(mylist2)
```

Sorted

demo.py > ...

```python
mylist = [99,20,38,10,80]

print(sorted(mylist))

print(sorted(mylist,reverse=True))
```

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```
D:\code\python_code\python_2_code>python demo.py
[10, 20, 38, 80, 99]
[99, 80, 38, 20, 10]
```

Max and Min

```
mylist = [99,20,38,10,80]

print(max(mylist))

print(min(mylist))
```

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```
D:\code\python_code\python_2_code>python demo.py
99
10
```

Sum

demo.py > ...

```python
mylist = [10,20,30,40,50]

print(sum(mylist))

```

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```
D:\code\python_code\python_2_code>python demo.py
150
```

count()

- The count() method returns the number of elements with the specified value.

demo.py > ...

```python
mylist1 = ["Java","Python",10,"Angular",99,"Python"]

mycount = mylist1.count("Python")
print(mycount)
```

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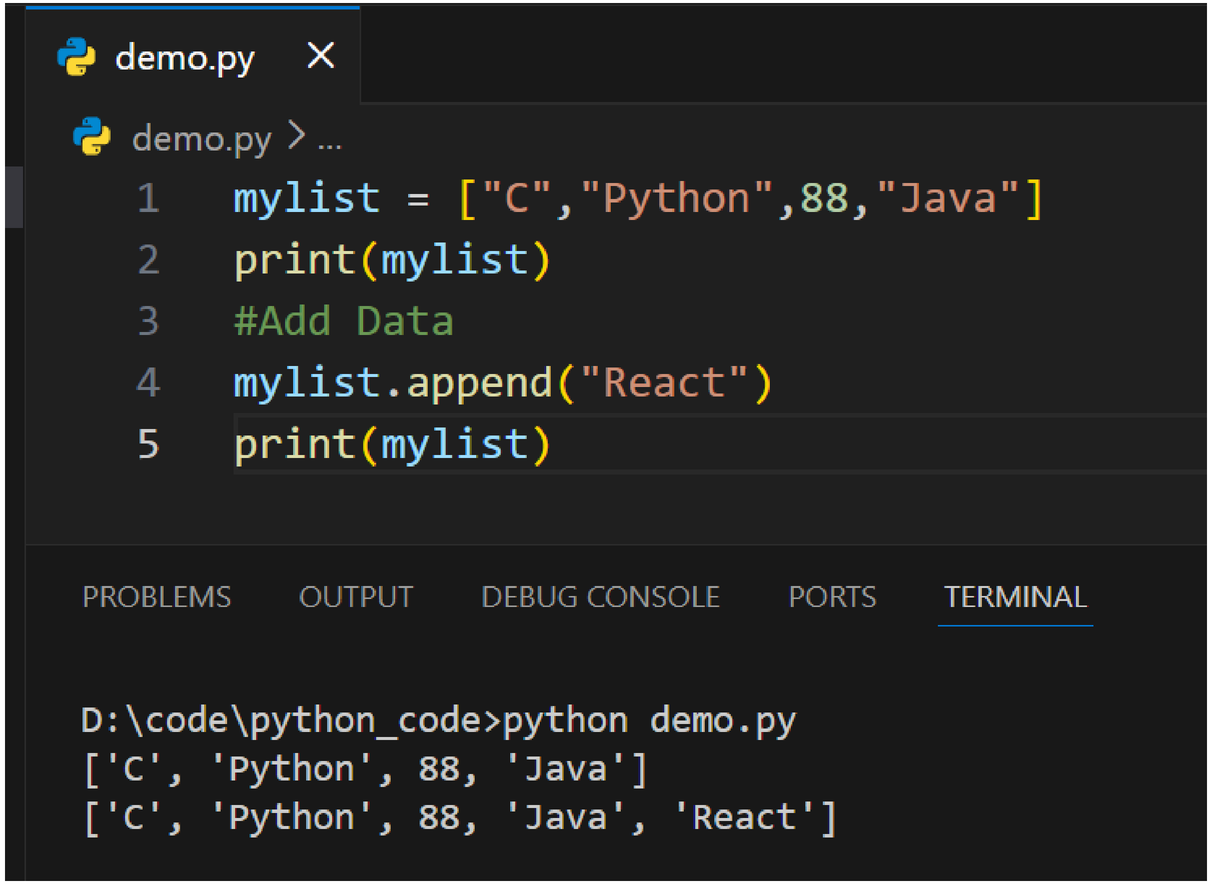
```
D:\code\python_code>python demo.py
2
```

append()

- Append is used to add or append a new element to a list. Only one element can be appended to a list. Append does not return any values, and appending is done in-place.




```
mylist = ["C","Python",88,"Java"]
print(mylist)
#Add Data
mylist.append("React")
print(mylist)
```

insert()

- The insert() method inserts the specified value at the specified position.

demo.py > ...

```python
mylist1 = ["Java","Python",10,"Angular"]

mylist1.insert(2,"Django")
print(mylist1)
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL

```
D:\code\python_code>python demo.py
['Java', 'Python', 'Django', 10, 'Angular']
```

index()

- The index() method returns the position at the first occurrence of the specified value.

demo.py > ...

```python
mylist1 = ["Java","Python",10,"Angular",99,"Python"]

position = mylist1.index("Python")
print(position)
```

PROBLEMS   OUTPUT   DEBUG CONSOLE   PORTS   TERMINAL

```
D:\code\python_code>python demo.py
1
```

pop()

- The pop() method removes the element at the specified position.

demo.py > ...

```python
mylist1 = ["Java","Python",10,"Angular"]

mylist1.pop(1)
print(mylist1)
```

PROBLEMS   OUTPUT   DEBUG CONSOLE   PORTS   TERMINAL

```
D:\code\python_code>python demo.py
['Java', 10, 'Angular']
```

remove()

- The remove() method removes the first occurrence of the element with the specified value.

demo.py > ...

```python
mylist1 = ["Java","Python",10,"Angular","Python"]

mylist1.remove("Python")
print(mylist1)

```

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```
D:\code\python_code>python demo.py
['Java', 10, 'Angular', 'Python']
```

clear()

- The clear() method removes all the elements from a list.

demo.py

demo.py > ...

```python
mylist = ["C","Python",88,"Java"]
print(mylist)
#Remove All Data
mylist.clear()
print(mylist)
```

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```
D:\code\python_code>python demo.py
['C', 'Python', 88, 'Java']
[]
```

```python
mylist = ["C","Python",88,"Java"]
print(mylist)
#Remove All Data
mylist.clear()
print(mylist)
```

copy()

- The copy() method returns a copy of the specified list.

demo.py

demo.py > ...

```python
mylist = ["C","Python",88,"Java"]
print(mylist)
#Copy All Data
mynewlist = mylist.copy()
print(mynewlist)
```

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```
D:\code\python_code>python demo.py
['C', 'Python', 88, 'Java']
['C', 'Python', 88, 'Java']
```


```python
mylist = ["C","Python",88,"Java"]
print(mylist)
#Copy All Data
mynewlist = mylist.copy()
print(mynewlist)
```

extend()

- The extend() method adds the specified list elements (or any iterable) to the end of the current list.

demo.py > ...

```python
mylist1 = ["Java","Python",10,"Angular"]
mylist2 = ["Mysql","MongoDB"]

mylist1.extend(mylist2)
print(mylist1)
```

PROBLEMS   OUTPUT   DEBUG CONSOLE   PORTS   TERMINAL

```
D:\code\python_code>python demo.py
['Java', 'Python', 10, 'Angular', 'Mysql', 'MongoDB']
```


```python
mylist1 = ["Java","Python",10,"Angular"]
mylist2 = ["Mysql","MongoDB"]

mylist1.extend(mylist2)
print(mylist1)
```

Tuple

Python Tuple Methods

- Python has two built-in methods that you can use on tuples.

- count
- index

count()

- The count() method returns the number of times a specified value appears in the tuple.

demo.py > ...

```python
mytuple = ("Java","Python",10,"Angular","Python")

ans = mytuple.count("Python")
print(ans)

```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL

```
D:\code\python_code>python demo.py
2
```

index()

- The index() method finds the first occurrence of the specified value.

- The index() method raises an exception if the value is not found.

Example

demo.py > ...

```python
mytuple = ("Java","Python",10,"Angular","Python")

ans = mytuple.index("Python")
print(ans)
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL

```
D:\code\python_code>python demo.py
1
```

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